

Smart Imaging

☐ Use multi-ch-acquisition

Time lapse

☒ T acquisition

Time duration (min):

600

Time interval (min):

20

Z-stack

☐ Z acquisition

relative Z start (µm):

0

Set

relative Z end (µm):

0

Set

Z step (µm):

1

Channel Group

Channels

C1	None	exposure time (ms)	200
C2	None	exposure time (ms)	200
C3	None	exposure time (ms)	100
C4	None	exposure time (ms)	200

File Directory:

D:\User Data\flow20240618_overnight

File Name:

frn_as

Chan to find cell

Cell numbers

20

Acquire

Abort

Segmentation

diameter (pixels):

30

calibrate

chan to segment:

Grayscale

Segmentation model

cyto3

☐ Load GPU

Run Segmentation

9 ROIs

additional settings

Analysis

☒ Real time analysis

Analysis instruction

segment nucleus in first channel, detect lysosomes in second and third channel, ca...

☐ Use language

Custom analysis code

LLM Python script

LLM endpoint

model

LLM API key

API version

Save

puncta threshold

0.2

co-distance threshold(pix)

42

Detect structure in chan:

Channel 4

Detect structure in chan (optional):

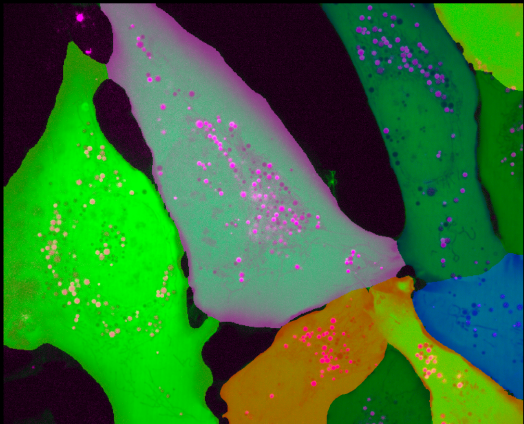
None

Detect structure in chan (optional):

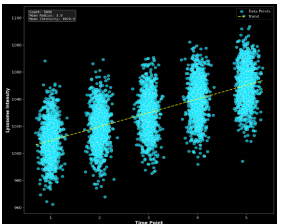
None

nuclear chan for nuclear distance (optional):

None



Real-time Analysis



Co-scientist

Co-Scientist

Experimental Data Analysis and Findings

Experimental Data Analysis and Findings

Algorithm: The primary goal of this experiment is to study the effect of optimal co-scientist in 2025, with a specific focus on changes in lysosome size and associated parameters. Key metrics analyzed include lysosome volume, intensity, and number in two conditions: optimal lysosome volume control (200).

Key Findings

- 1. Lysosome Volume: Lysosome volume is significantly smaller and more homogeneous in size as indicated by the lower standard deviation and narrower range (2000 - 30000).
- 2. Lysosome Intensity: Lysosome intensity is significantly higher and more homogeneous in size as indicated by the lower standard deviation and narrower range (10000 - 15000).
- 3. Lysosome Number: Lysosome number is significantly higher and more homogeneous in size as indicated by the lower standard deviation and narrower range (10000 - 15000).

Recommendation

Based on the findings, it is recommended to continue the use of optimal co-scientist in 2025, with a specific focus on changes in lysosome size and associated parameters. Key metrics analyzed include lysosome volume, intensity, and number in two conditions: optimal lysosome volume control (200).

Conclusion & Hypothesis

Based on the findings, it is recommended to continue the use of optimal co-scientist in 2025, with a specific focus on changes in lysosome size and associated parameters. Key metrics analyzed include lysosome volume, intensity, and number in two conditions: optimal lysosome volume control (200).